Technology Stack

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*[HTML, CSS, JavaScript]*	*[Renders the web UI for the Flask app. Lightweight and compatible with all browsers; no additional framework needed for simple form-based input and result display.]*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*[Python, Flask]*	*[Flask serves the web application and exposes REST API endpoints for model inference. Python is the primary language for ML development, ensuring seamless integration.]*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*[CSV / IBM Cloud Object Storage]*	*[Historical flood datasets are stored as CSV files. IBM Cloud Object Storage is used for persisting trained model files (.pkl) and datasets in the cloud environment.]*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*[IBM Cloud (Cloud Foundry)]*	*[IBM Cloud provides scalable, reliable hosting for the Flask app. Cloud Foundry enables easy deployment with minimal configuration, making the app accessible from any location.]*
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	*[GitHub]*	*[GitHub is used for source code versioning and team collaboration. Enables tracking of changes across ML experiments and web app development throughout the project lifecycle.]*

6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*[Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Jupyter Notebook]*	*[Scikit-learn and XGBoost are used for training ML models. Pandas and NumPy handle data processing. Matplotlib visualises results. Jupyter Notebook is used for model development and evaluation.]*